

General Chemistry (Chem1006)

Chapter 7

Equilibrium Concepts and Acid-base Equilibrium

Introduction

- Writing chemical equations involves placing reactant formulas on the left side of a reaction arrow and product formulas on the right side.
- **Reversible reactions**: Reactions that proceed in both forward (left to right) and reverse (right to left) directions.

➢ They are represented by a **double arrow** (⇌).

- **Chemical equilibrium:** A state where the forward and reverse reactions occur at equal rates and the concentrations of products and reactants remain constant.

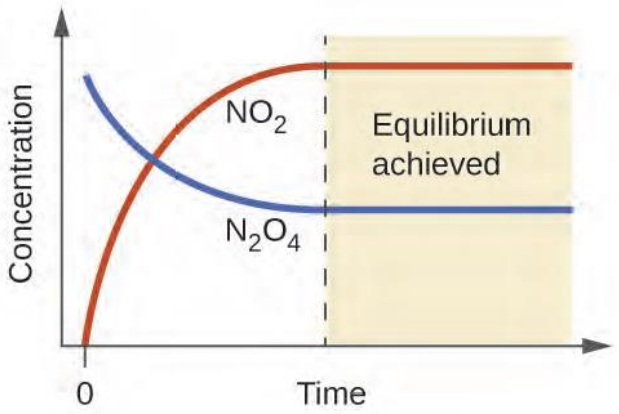
➢ In a chemical equilibrium, the forward and reverse reactions do not stop, rather they continue to occur at the same rate, hence,

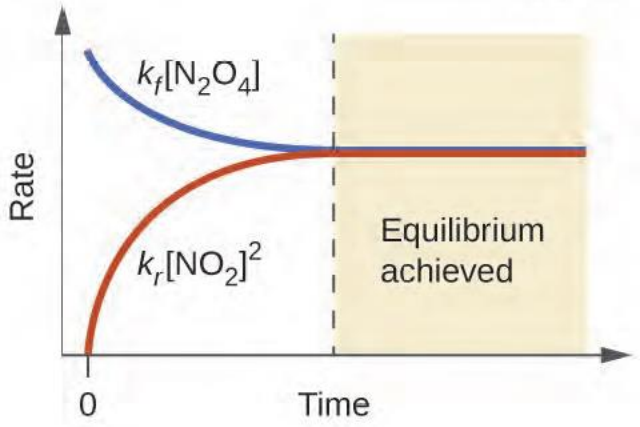
✓ Chemical equilibrium is a dynamic process

Example 1: Consider the decomposition of dinitrogen tetraoxide (colorless) to nitrogen dioxide (brown) in a closed vessel.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

- N_2O_4 decompose to form NO_2, & the NO_2 produced reacts to form N_2O_4.
- As soon as the forward reaction produces any NO_2, the reverse reaction begins and NO_2 starts to react to form N_2O_4.
- At equilibrium, the concentrations of N_2O_4 and NO_2 no longer change because
  - ✓ the rate of formation of NO_2 is exactly equal to the rate of consumption of NO_2, and
  - ✓ the rate of formation of N_2O_4 is exactly equal to the rate of consumption of N_2O_4.





7.1. Chemical Equilibrium

- The status of a reversible reaction is conveniently assessed by evaluating its reaction quotient (Q).
- For a reversible reaction described by

$$mA + nB \rightleftharpoons xC + yD$$

- The reaction quotient is derived directly from the stoichiometry of the balanced equation as

$$Q_c = \frac{[C]^x[D]^y}{[A]^m[B]^n}$$

- The reaction quotient is equal to the molar concentrations of the products of the chemical equation (multiplied together) over the reactants (also multiplied together), with each concentration raised to the power of the coefficient of that substance in the balanced chemical equation.
- If the reactants and products are gaseous, a reaction quotient may be similarly derived using partial pressures:

$$Q_p = \frac{P_C^{x} P_D^{y}}{P_A^{m} P_B^{n}}$$

Note: Disregard units when computing Q

- Q is dimensionless

Example 2: Write the concentration-based and pressure-based reaction quotient expression for each of the following reaction:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

Answer:

$$Q_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]} \qquad Q_p = \frac{P_{SO_3}{}^2}{P_{SO_2}{}^2 P_{O_2}}$$

- As the reaction proceeds toward equilibrium in the forward direction, **reactant concentrations decrease, product concentration increases and the reaction quotient consequently increases**.
- **When equilibrium is achieved, the concentrations of reactants and product remain constant, as does the value of Qc**.

- **Equilibrium constant, K:** The constant value of Q exhibited by a system at equilibrium

$$Q_c \text{ at equilibrium} = K_c = \frac{[C]^x[D]^y\ldots}{[A]^m[B]^n\ldots}$$

➢ This equation is a mathematical statement of the law of mass action:

Law of mass action: When a reaction has attained equilibrium at a given temperature, **the reaction quotient for the reaction always has the same value**.

Note: Solids, pure liquids and solvents do not appear in the equilibrium constant expressions (their activities are considered to be 1)

Example 3: For the reaction: **$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$,** the concentrations at equilibrium are $[SO_2]$ = 0.90 M, $[O_2]$ = 0.35 M, and $[SO_3]$ = 1.1 M.

What is the value of the

a) reaction quotient before any reaction occurs?

b) equilibrium constant, Kc?

Solution

a) $Q_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$

Before the reaction starts, $[SO_3] = 0$

Therefore, Qc = 0

b) $K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]} = \frac{(1.1)^2}{(0.90)^2(0.35)} = 4.3$

- The magnitude of equilibrium constant explicitly reflects the composition of a reaction mixture at equilibrium

- A reaction exhibiting **a large K** will **reach equilibrium when most of the reactant has been converted to product**
- A **small K** indicates **the reaction achieves equilibrium after very little reactant has been converted**.
- The magnitude of **K does not indicate how rapidly or slowly equilibrium will be reached**
- Some equilibria are established so quickly and others so slowly

Predicting the Direction of Reaction

Example 4: Consider the reaction:

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g) \qquad Kc = 0.64$$

Determine in which direction the reaction proceeds as it goes to equilibrium in each of the three experiments shown.

Reactants/Products	Experiment 1	Experiment 2
$[CO]_i$	0.020 M	0.011 M
$[H_2O]_i$	0.020 M	0.0011 M
$[CO_2]_i$	0.0040 M	0.037 M
$[H_2]_i$	0.0040 M	0.046 M

Solution: Experiment 1

$$Qc = \frac{[CO_2][H_2]}{[CO][H_2O]} = \frac{(0.004)(0.004)}{(0.020)(0.020)} = 0.04$$

Qc < Kc

Therefore, the reaction **will proceed in the forward direction**

Experiment 2

$$Qc = \frac{(0.037)(0.046)}{(0.011)(0.0011)} = 1.4 \times 10^2$$

Qc > Kc

Therefore, the reaction **will proceed in the reverse direction**

Example 5: Calculate the reaction quotient & determine the direction in which each of the following reactions will proceed to reach equilibrium.

a) A 1.00 L flask containing 0.0500 mol of NO(g), 0.0155 mol of Cl_2(g), & 0.500 mol of NOCl:

2NO(*g*) + Cl_2(*g*) ⇌ 2NOCl(*g*) $Kc = 4.6 \times 10^4$

$$Qc = \frac{[NOCl]^2}{[NO]^2[Cl_2]} = \frac{(0.500)^2}{(0.0500)^2(0.0155)} = 6.45 \times 10^3$$

- The **reaction shifts to the right**

b) A 5.0 L flask containing 17 g of NH_3, 14 g of N_2, and 12 g of H_2:

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ *Kc* = 0.060 (Molar masses: NH_3 = 17, N_2 = 28, H_2 = 2)

Solution: $M = \frac{n}{V(L)} = \frac{actual\ mass}{molar\ mass \times V(L)}$

$\mathrm{M}\ (NH_3) = 0.2$, $\mathrm{M}\ (N_2) = 0.1$, $\mathrm{M}(H_2) = 1.2$

$$Qc = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(0.2)^2}{(0.1)(1.2)^3} = 0.23$$

➢ The **reaction shifts to the left**.

c) A 2.00 L flask containing 230 g of SO_3(g):

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ *Kc* = 0.230

$[SO_2] = 0$

Therefore, Kc = 0

➢ The **reaction shifts to the right**

Homogeneous Equilibria

Homogeneous equilibrium: The one in which **all of the reactants and products are present in a single solution**

Most common types of homogeneous equilibria:

a) those occurring **in liquid-phase solutions** and

b) those involving **exclusively gaseous species**.

- The chemical species involved can be molecules, ions, or a mixture of both.

Examples:

a) $C_2H_2(aq) + 2Br_2(aq) \rightleftharpoons C_2H_2Br_4(aq)$ $\qquad K_c = \dfrac{[C_2H_2Br_4]}{[C_2H_2][Br_2]^2}$

$HF(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + F^-(aq)$ $\qquad K_c = \dfrac{[H_3O^+][F^-]}{[HF]}$

$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$ $\qquad K_c = \dfrac{[NH_4^+][OH^-]}{[NH_3]}$

- Since **$H_2O(l)$ is the solvent for these solutions, its concentration does not appear as a term in the *Kc* expression**

b) $3O_2(g) \rightleftharpoons 2O_3(g)$ $\qquad K_c = \dfrac{[O_3]^2}{[O_2]^3}$

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $\qquad K_c = \dfrac{[NH_3]^2}{[N_2][H_2]^3}$

$C_3H_8(g) + 5O_2(g) \rightleftharpoons 3CO_2(g) + 4H_2O(g)$ $\qquad K_c = \dfrac{[CO_2]^3[H_2O]^4}{[C_3H_8][O_2]^5}$

Note: The **concentration of $H_2O(g)$ has been included in the last example because water is not the solvent in this gas-phase reaction and its concentration changes.**

Whenever gases are involved in a reaction, **the partial pressure of each gas can be used instead of its concentration in the equilibrium expression**.

Reason: The partial pressure of a gas is **directly proportional to its concentration at constant temperature**.

$$PV = nRT \qquad P = \left(\frac{n}{V}\right)RT = MRT$$

Example: For the reaction:

$$C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$$

$$Q_P = \frac{P_{C_2H_4} P_{H_2}}{P_{C_2H_6}}$$

At equilibrium,

$$K_P = Q_P = \frac{P_{C_2H_4} P_{H_2}}{P_{C_2H_6}}$$

The relationship between Kc and K_P

For the gas-phase reaction ***m*A + *n*B ⇌ *x*C + *y*D**

$$K_P = \frac{(P_C)^x (P_D)^y}{(P_A)^m (P_B)^n}$$

$$= \frac{([\mathrm{C}] \times RT)^x ([\mathrm{D}] \times RT)^y}{([\mathrm{A}] \times RT)^m ([\mathrm{B}] \times RT)^n}$$

$$= \frac{[\mathrm{C}]^x [\mathrm{D}]^y}{[\mathrm{A}]^m [\mathrm{B}]^n} \times \frac{(RT)^{x+y}}{(RT)^{m+n}}$$

$$= K_c (RT)^{(x+y)-(m+n)}$$

$$= K_c (RT)^{\Delta n}$$

Hence,

$K_P = Kc(RT)^{\Delta ng}$

Example: Write the equations for the conversion of *Kc* to K_P for each of the following reactions, which occur in the gas phase:

a) $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$

b) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

c) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

d) At 227 °C, the following reaction has *Kc* = 0.0952:

$$\mathbf{CH_3OH(g) \rightleftharpoons CO(g) + 2H_2(g)}$$

What would be the value of K_P at this temperature? (R = 0.0821 L atm $K^{-1}mol^{-1}$)

Solution

a) $\Delta n_g = (1 + 1) - (1 + 1) = 0$

$K_P = Kc(RT)^{\Delta ng} = Kc\ (RT)^0$

$\mathbf{K_P = Kc}$

Note: If $\Delta n_g = 0$, $K_P = Kc$

b) $\Delta n_g = (2) - (2 + 1) = -1$ $K_P = Kc(RT)^{\Delta ng}$ $\mathbf{K_P = Kc\ (RT)^{-1}}$

c) $\Delta n_g = 2 - 1 = 1$ $\mathbf{K_P = Kc\ (RT)}$

d) $\Delta n_g = (2 + 1) - 1 = 2$

$K_P = Kc\ (RT)^2$

$K_P = 0.0952 \times (0.0821 \times 500.15)^2 = 160 = \mathbf{1.6 \times 10^2}$

Heterogeneous Equilibria

Heterogeneous equilibrium: It is a system in which **reactants and products are found in two or more phases**.

- Heterogeneous equilibria also involve phase changes

Examples:

1. $PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2Cl^-(aq)$ $\quad Kc = [Pb^{2+}][Cl^-]^2$

2. $CaO(s) + CO_2(g) \rightleftharpoons CaCO_3(s)$ $\quad Kc = \frac{1}{[CO_2]}$, $K_P = \frac{1}{P_{CO_2}}$

3. $Br_2(l) \rightleftharpoons Br_2(g)$ $\quad Kc = [Br_2]$

7.2. Le Chatelier's Principle

Le Chatelier's principle: When a chemical system at equilibrium is disturbed, **it returns to equilibrium by counteracting the disturbance**

➢If a system at equilibrium is subjected to a perturbance or stress (such as a change in concentration), **the position of equilibrium changes to a direction that avoids the stress**.

- ✓Since this **stress affects the concentrations of the reactants and the products**, the value of ***Q* will no longer be equal the value of *K*.**
- ✓To re-establish equilibrium, **the system will either shift toward the products** (if $Q < K$) or **the reactants** (if $Q > K$) until ***Q* returns to the same value as *K***.

Predicting the Direction of a Reversible Reaction

1. Effect of Change in Concentration on Equilibrium

- A chemical system at equilibrium can be temporarily shifted out of equilibrium **by adding or removing one or more of the reactants or products.**

➢ The concentrations of both reactants and products then undergo additional changes to return the system to equilibrium.

Consider the following reaction to be at equilibrium:

$$\mathbf{Fe^{3+}(\mathit{aq}) + SCN^{-}(\mathit{aq}) \rightleftharpoons Fe(SCN)^{2+}(\mathit{aq})}$$

What will happen to the equilibrium position if **silver nitrate is added** to the solution?

Answer: Addition of silver nitrate precipitates some of the SCN^- as the white solid AgSCN.

$$Ag^+(aq) + SCN^-(aq) \rightleftharpoons AgSCN(s).$$

The **decrease in the SCN^- concentration shifts the first equilibrium in the solution to the left**, decreasing the concentration of the $Fe(SCN)^{2+}$.

Example: A mixture of gases at 400 °C with $[H_2]$= $[I_2]$ = 0.221 *M* and [HI] = 1.563 *M* is at equilibrium; for this mixture, *Qc* = *Kc* = 50.0

$$\mathbf{H_2(}g\mathbf{) + I_2(}g\mathbf{) \rightleftharpoons 2HI(}g\mathbf{)}$$

If H_2 is introduced into the system so quickly that its concentration doubles before it begins to react (new $[H_2]$ = 0.442 *M*), what will be the concentration of the species when an equilibrium is re-established?

Answer

- ✓ The **equilibrium constant remains** the same.
- ✓ The **stress is relieved when the reaction shifts to the right, using up some (but not all) of the excess H_2, reducing the amount of uncombined I_2, and forming additional HI**.

2. Effect of Change in Pressure on Equilibrium

- Changes in pressure have a measurable effect only in systems in which

i. gases are involved, and

ii. the chemical reaction produces a change in the total number of gas molecules

✓ We should have different numbers of moles of gas on the reactant and product sides

N.B.: Addition of a gas not involved in the equilibrium will not perturb the equilibrium

As we **increase the pressure of a gaseous system at equilibrium, either by decreasing the volume of the system or by adding more of one of the components of the equilibrium mixture, we introduce a stress by increasing the partial pressures** of one or more of the components.

➢ A shift in the equilibrium that reduces the total number of molecules per unit of volume **will be favored because this relieves the stress**.

➢ The reverse reaction would be favored by a decrease in pressure.

Example 1: What happens when we increase the pressure on a system in which NO, O_2, & NO_2 are at equilibrium in the following reaction?

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

Answer:

- The formation of additional amounts of NO_2 decreases the total number of molecules in the system because each time **two molecules of NO_2 form, a total of three molecules of NO and O_2 are consumed**.
- This **reduces the total pressure exerted by the system and reduces**, but does not completely relieve, the stress of the increased pressure.
- As a result the **equilibrium shifts to the right**.
- The amount of **NO and O_2 decreases while that of NO_2 increases**
- On the other hand, a decrease in the pressure on the system **favors decomposition of NO_2 into NO and O_2**, which tends to restore the pressure.

Example 2: Consider the following reaction at equilibrium:

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$

- Because there is no change in the total number of molecules in the system during reaction, **a change in pressure does not favor either formation or decomposition of gaseous nitrogen monoxide**.
- ✓ A change in pressure **will not disturb the equilibrium position**

3. Effect of Change in Temperature on Equilibrium

- **Changing concentration or pressure perturbs an equilibrium** because the reaction quotient is shifted away from the equilibrium value.
- **Changing the temperature of a system at equilibrium actually changes the value of the equilibrium constant**.

Example: Consider the following reaction:

$$\mathbf{H_2(}g\mathbf{) + I_2(}g\mathbf{) \rightleftharpoons 2HI(}g\mathbf{)} \qquad \Delta H = -9.4 \text{ kJ}$$

- Because this reaction is exothermic, we can write it with heat as a product.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) + \text{heat}$$

- Increasing the temperature of the reaction **increases the internal energy of the system.**
- The **reaction shifts to the left to relieve the stress**,
- There is **an increase in the concentration of H_2 and I_2 and a reduction in the concentration of HI**.
- The **equilibrium constant decreases**

- Lowering the temperature of the system **reduces the amount of energy present, and hence favors the production of heat**,

✓ favors the formation of **more of hydrogen iodide**

✓ The **equilibrium constant increases**

- Consider the following reaction at equilibrium:

$$\mathbf{N_2O_4(}g\mathbf{) \rightleftharpoons 2NO_2(}g\mathbf{)} \qquad \Delta H = 57.20 \text{ kJ}$$

➢ The positive ΔH value tells us that the reaction is endothermic and could be written as

$$N_2O_4(g) + \text{heat} \rightleftharpoons 2NO_2(g)$$

➢ Increasing the temperatures (adding energy), the **equilibrium shifts to the right**

✓ more amount of the brown NO_2 is formed

- If, however, we put a stress on the system by **cooling the mixture (withdrawing energy), the equilibrium shifts to the left to supply some of the energy lost by cooling**.
- ✓ The concentration of **colorless N_2O_4 increases**, and the concentration of **brown NO_2 decreases**, causing the brown color to fade.

N.B.:

- A **catalyst can speed up the rate of a reaction**.
  - ✓ It may **cause a system to reach equilibrium more quickly**
  - ✓ But, a catalyst **has no effect on the value of an equilibrium constant nor on equilibrium concentrations**.
- **Increasing temperature increases the speed of a reaction** (for both exothermic and endothermic).
  - ✓ Hence **equilibrium is reached fast**.
  - ✓ For exothermic reactions **increasing the temperature decreases the yield** and if the temperature is **decreased equilibrium is reached very slowly**.

- ✓ As a result, the **temperature has to be optimized**
- ✓ Part of the rate of formation **lost by operating at lower temperatures can be recovered by using a catalyst**.

- In the **commercial production of ammonia**, conditions of about 500°C, 150 – 900 atm, and the presence of a catalyst are used to give the best compromise among rate, yield, and the cost of the equipment

$$\mathbf{N_2(}g\mathbf{) + 3H_2(}g\mathbf{) \longrightarrow 2NH_3(}g\mathbf{)} \qquad \Delta H = -92.2 \text{ kJ}$$

7.3. Equilibrium Calculations

- Changes in concentrations or pressures of reactants and products occur as a reaction system approaches equilibrium.

➢ These changes can be related to each other using **the coefficients in the balanced chemical equation describing the system**

Example: Consider the decomposition of ammonia on heating in a closed system.

$$\mathbf{2NH_3(\mathit{g}) \rightleftharpoons N_2(\mathit{g}) + 3H_2(\mathit{g})}$$

➢ If the concentration of N_2 increases by 0.11 *M*,

$\Delta[N_2] = 0.11$ M,

$\Delta[H_2] = 3\times \Delta[N_2] = 3 \times 0.11$ M $= 0.33$ M

$\Delta[NH_3] = -2\times \Delta[N_2] = -2 \times 0.11$ M $= -0.22$ M

➢ **In general, if $\Delta[N_2] = x$, $\Delta[H_2] = 3x$, $\Delta[NH_3] = -2x$**

The coefficients in the Δ terms are identical to those in the balanced equation for the reaction.

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$$
$$-2x \qquad x \qquad 3x$$

Example: Complete the changes in concentrations for each of the following reactions:

a) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

_____ x _____

b) $C_4H_8(g) \rightleftharpoons 2C_2H_4(g)$

_____ $-2x$

c) $4NH_3(g) + 7H_2O(g) \rightleftharpoons 4NO_2(g) + 6H_2O(g)$

_____ _____ _____ _____

Answer:

a) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

2x x **-2x**

b) $C_4H_8(g) \rightleftharpoons 2C_2H_4(g)$

x $-2x$

c) $4NH_3(g) + 7H_2O(g) \rightleftharpoons 4NO_2(g) + 6H_2O(g)$

-4x **-7x** **4x** **6x**

Calculations Involving Equilibrium Concentrations

- The value of the reaction quotient , *Qc,* of any reaction at equilibrium is equal to its equilibrium constant, Kc. (Qc = Kc)
- The mathematical expression for *Qc can be used* to determine a number of quantities associated with a reaction at equilibrium.
- There are only three basic types of calculations:

1. **Calculation of an equilibrium constant:** If concentrations of reactants & products at equilibrium are known, the value **of Kc can be calculated**.
2. **Calculation of missing equilibrium concentrations:** If Kc & all of the equilibrium concentrations, except one, are known, t**he remaining concentration can be calculated.**
3. **Calculation of equilibrium concentrations from initial concentrations:** If **Kc & a set of concentrations of reactants & products that are not at equilibrium are known**, the concentrations at equilibrium can be calculated.

1. Calculation of an Equilibrium Constant

Consider the following reversible reaction:

$$I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq)$$

If a solution with the concentrations of I_2 and I^- both equal to 1.000×10^{-3} *M* before reaction gives an equilibrium concentration of I_2 of 6.61×10^{-4} *M*, what is the equilibrium constant for the reaction?

Solution:

Steps:

Calculating the changes in concentration, determine the equilibrium concentrations and calculate the equilibrium constant

Initial concentrations (M): $I_2 = 1.000 \times 10^{-3}$, $I^- = 1.000 \times 10^{-3}$, $I_3^- = 0$

Change (M): $I_2 = -x$ $I^- = -x$, $I_3^- = +x$

Equilibrium concentrations (M): $I_2 = 1.000 \times 10^{-3} - x$, $I^- = 1.000 \times 10^{-3} - x$, $I_3^- = x$

Equilibrium concentration of I_2 (M)= $\mathbf{6.61 \times 10^{-4}}$

- $1.000 \times 10^{-3} - x = 6.61 \times 10^{-4}\ M$
- $X = 3.39 \times 10^{-4}\ M$

equilibrium concentration of I^- (M) = $1.000 \times 10^{-3} - x$

$= 1.000 \times 10^{-3} - 3.39 \times 10^{-4}$

$= \mathbf{6.61 \times 10^{-4}\ M}$

equilibrium concentration of I_3^-(M) = x $= 3.39 \times 10^{-4}$

$$Kc = Qc = \frac{[I_3^-]}{[I_2][I^-]} = \frac{3.39 \times 10^{-4}}{6.61 \times 10^{-4} \times 6.61 \times 10^{-4}} = 776$$

2. Calculation of missing equilibrium concentrations

Example 1: The equilibrium constant for the reaction of nitrogen & hydrogen to produce ammonia at a certain temperature is 6.00×10^{-2}. Calculate the equilibrium concentration of ammonia if the equilibrium concentrations of nitrogen and hydrogen are 4.26 *M* and 2.09 *M*, respectively.

Solution:

Chemical equation: $\mathbf{N_2\,(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)}$

$$\boldsymbol{Kc} = \frac{[\mathbf{NH_3}]^2}{[\mathbf{N_2}][\mathbf{H_2}]^3}$$

$$[\mathrm{NH_3}] = \sqrt{Kc[\mathrm{N_2}][\mathrm{H_2}]^3} = \sqrt{6.00 \times 10^{-2}\, x\, 4.26\, x\, (2.09)^3} = 1.53\ M$$

3. Calculation of changes in concentration and equilibrium concentrations from initial concentrations

Steps:

1. Determine the direction the reaction proceeds to come to equilibrium.
2. Determine the relative changes needed to reach equilibrium, then write the equilibrium concentrations in terms of these changes.
3. Solve for the change and the equilibrium concentrations.

Example: The equilibrium constant for the decomposition of $PCl_5(g)$ into $PCl_3(g)$ & $Cl_2(g)$ is 0.0211. What are the equilibrium concentrations of PCl_5, PCl_3, and Cl_2 if the initial concentration of PCl_5 was 1.00 *M*?

Solution: Chemical equation:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

We have no products initially,

- $Qc = 0$ and
- the reaction will proceed to the right.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

Change in concentration: $PCl_5 = -x$, $PCl_3 = x$, $Cl_2 = x$

***Equilibrium concentrations** (M):* $PCl_5 = 1.00 - x$, $PCl_3 = x$, $Cl_2 = x$

$$Kc = \frac{[PCl_3][Cl_2]}{PCl_5} = \frac{x^2}{1.00-x} = 0.0211$$

$x^2 + 0.0211x - 0.0211 = 0$, x = 0.135, or x = − 0.156

Note: x can not be negative nor greater than the initial amount

Take the positive solution. **x = 0.135**

The equilibrium concentrations are:

$[PCl_5] = 1.00 - 0.135 = 0.87\ M$, $[PCl_3] = x = 0.135\ M$, $[Cl_2] = x = 0.135\ M$

Approximate Solutions:

- If **Kc is very small**, the amount reacted at equilibrium is **negligible when compared with the reactant**.
- If **Kc is very large**, the amount of the **limiting reactant at equilibrium is negligible**

Example: What are the concentrations at equilibrium of a 0.15 *M* solution of HCN?

$HCN(aq) \rightleftharpoons H^+(aq) + CN^-(aq) \qquad Kc = 4.9 \times 10^{-10}$

Solution:

Equilibrium concentrations: $H^+ = x$, $CN^- = x$, $HCN = 0.15 - x$

$$4.9 \times 10^{-10} = \frac{[H^+][CN^-]}{[HCN]} = \frac{x^2}{0.15-x}$$

$[H^+] = [CN^-] = x = \mathbf{8.6 \times 10^{-6}}$, $[HCN] = 0.15 - 8.6 \times 10^{-6} = \mathbf{0.15}$

Approximation:

Kc is very small

- X << 0.15
- **0.15 – x ≈ 0.15**
- The concentration of HCN nearly remains constant

$$4.9 \times 10^{-10} = \frac{[H^+][CN^-]}{[HCN]} = \frac{x^2}{0.15}$$

$x = 8.6 \times 10^{-6}$

- **Solving the exact (quadratic) equation and using approximations gave the same result to two significant figures.**
- The error introduced is $\frac{8.6x10^{-6}}{0.15} x100 = \mathbf{0.006}\%$
- The **error introduced is less than 5%**
- The **approximation is valid**

7.4. Concepts of acid-base

7.4.1. Arrhenius concept

Acid: A compound that dissolves in water to yield **hydronium ions** (H_3O^+)

Base: A compound that dissolves in water to yield **hydroxide ions** (OH^-).

Limitations:

The Arrhenius definitions of **acidity and alkalinity are restricted to aqueous solutions** and **refer to the concentration of the solvated ions**.

Examples:

- ✓ Pure H_2SO_4 or HCl dissolved in toluene **are not acidic even if both can donate a proton to toluene.**
- ✓ A solution of sodium amide ($NaNH_2$) in liquid ammonia is **not alkaline even if the amide ion (NH_2^-) can readily deprotonate ammonia**.

7.4.2. BrØnsted-Lowery concept:

Bronsted-Lowry acid: A compound that **donates a proton to another compound Acids**:

- ➢ may be **neutral molecules** such as HCl & CH_3COOH; **anions** such as HSO_4^- & HCO_3^- and **cations** such as H_3O^+, NH_4^+, & $[Al(H_2O)_6]^{3+}$)

Bronsted-Lowry base: A compound that **accepts a proton**

- Bases may be **neutral molecules** such as NaOH, NH_3, & CH_3NH_2; **anions** such as OH^-, HS^-, HCO_3^-, F^-; or **cations** such as $[Al(H_2O)_5OH]^{2+}$

- The product that remains after an acid donates a proton is called the **conjugate base** of the acid.
  - This species is a base because it can accept a proton (to re-form the acid):

acid ⇌ proton + conjugate base

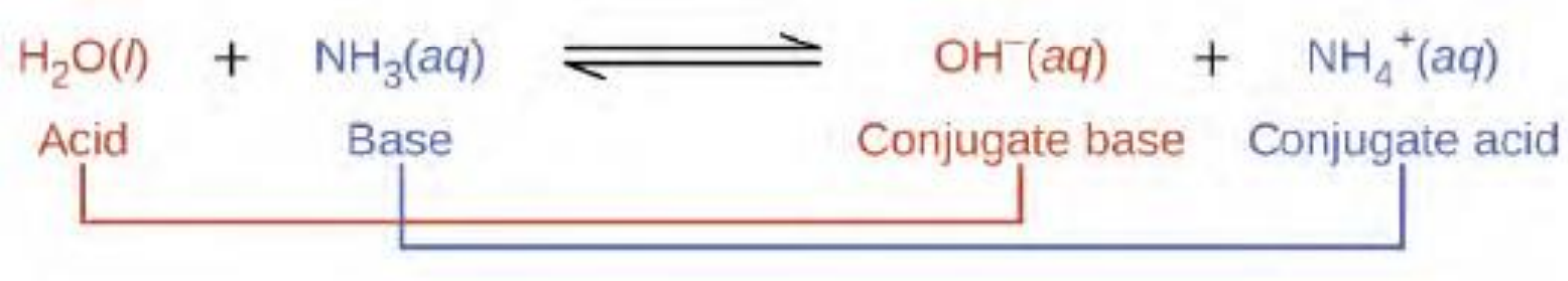
$HF \rightleftharpoons H^+ + F^-$

- The product that results when a base accepts a proton is called the **conjugate acid** of the base.
  - This species is an acid because it can give up a proton (and thus re-form the base):

base + proton ⇌ conjugate acid

$OH^- + H^+ \rightleftharpoons H_2O$ $\quad$ $H_2O + H^+ \rightleftharpoons H_3O^+$ $\quad$ $CO_3^{2-} + H^+ \rightleftharpoons HCO_3^-$

- Consider the acid-base reaction that takes place when ammonia is dissolved in water.
- A **water molecule (functioning as an acid**) transfers a proton to an ammonia molecule (**functioning as a base**), yielding the conjugate base of water, OH^-, and the conjugate acid of ammonia, NH_4^+.



- The reaction between a Brønsted-Lowry acid and water is called **acid ionization**.

Example:

$$H_2O + HF \rightleftharpoons H_3O^+ + F^-$$

base acid acid base

$$K = \frac{[H_3O^+][F^-]}{[HF]}$$

- When we add a base to water, a **base ionization** reaction occurs in which protons are transferred from water molecules to base molecules.

Example:



$$K = \frac{[C_5NH_6^+][OH^-]}{[C_5NH_5]}$$

Autoionization: A reaction, in which a substance ionizes when **one molecule of the substance reacts with another molecule of the same substance**,

Example:

Pure water undergoes autoionization

- The equilibrium constant for the ionization of water is called the **ion-product constant for water (*K*w)**:

$H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$ $\qquad$ $Kw = [H_3O^+][OH^-]$

- **At 25 °C, *K*w = 1.0 × 10^{-14}**
- **At 100 °C, *K*w = 1.0 × 10^{-13}**

Example : What are the H_3O^+ concentration and the OH^- concentration in pure water at 25 °C?

Solution

$$\boldsymbol{K}\mathbf{w} = [H_3O^+][OH^-] = [H_3O^+]^2 = [OH^-]^2 = 1.0 \times 10^{-14}$$

$$[H_3O^+] = [OH^-] = 1.0 \times 10^{-14} = 1.0 \times 10^{-7}\ \boldsymbol{M}$$

Example: A solution of CO_2 in water has a H_3O^+ concentration of 2.0×10^{-6} *M*. What is the concentration of OH^- at 25 °C?

Solution

$$\boldsymbol{K}\mathbf{w} = [H_3O^+][OH^-] = 1.0 \times 10^{-14}$$

$$OH^- = \frac{Kw}{[H_3O^+]} = \frac{1.0 \times 10^{-14}}{2.0 \times 10^{-6}} = 5.0 \times 10^{-9}\ \mathbf{M}$$

Amphiprotic (amphoteric) Species: Molecules & ions that may either gain or lose a proton under the appropriate conditions

Example: Write separate equations representing the reaction of HSO_3^-

(a) as an acid with OH^- (b) as a base with HI

Solution:

(a) $HSO_3^-(aq) + OH^-(aq) \rightleftharpoons SO_3^{2-}(aq) + H_2O(l)$

(b)) $HSO_3^-(aq) + HI(aq) \rightleftharpoons H_2SO_3(aq) + I^-(aq)$

7.4.3. Lewis concept

Lewis acid: A species (molecule or ion) that can **accept a pair of electrons** (**an electrophile**)

✓ It has **vacant orbitals**, or the **ability to form a vacant orbital**.

Examples: H^+, K^+, Mg^{2+}, Fe^{3+}, BF_3 , $AlCl_3$, Br_2.

Lewis base: A species (molecule or ion) that can **donate a pair of electrons** (**a nucleophile**)

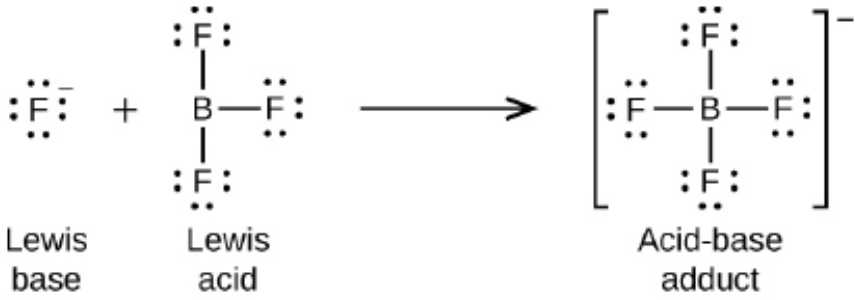
✓ It has lone pair electrons

Examples: OH^-, F^-, H_2O, ROH, NH_3, SO_4^{2-}, H^-, CO, C_6H_6

- In a **Lewis Acid-Base reaction a bond** is formed where both electrons come from the Lewis base, and this is called a **coordinate covalent bond** (or dative covalent bond).
- **There are three general types of Lewis Acids**

1. **Molecules with an incomplete octet**

Example: Boron Trifluoride

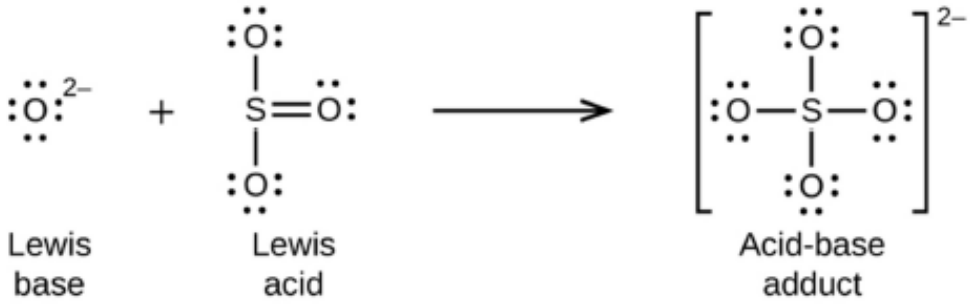


2. Molecules with double bonds

- A double bond has 4 electrons shared between 2 atoms in a π and σ (sigma) orbital.

- If the π bond is broken, the two electrons move to a **non-bonding orbital** on one atom, the other atom develops an incomplete octet and can function as a Lewis acid.

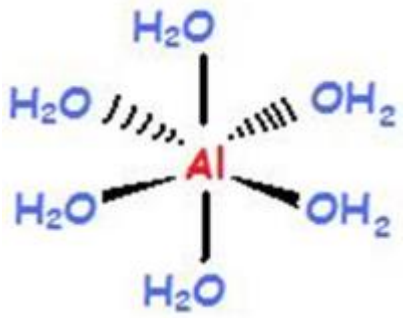
Example:



3. metal ions with vacant d-orbitals:

- Most metals react with water when they dissolve in water to form a complex ion.

Example: Al^{3+}



7.5. pH and pOH

- Hydronium and hydroxide ions are present both in pure water and in all aqueous solutions
- Their concentrations are inversely proportional as determined by the ion product of water (*K*w).

➢ A **neutral** solution contains equal concentrations of H_3O^+ and OH^-

➢ **An Acidic solution** contains a greater concentration of H_3O^+ than OH^-

➢ A **Basic solution** contains a lesser concentration of H_3O^+ than OH^-

- The scale for the concentration is based on the p-function

$$pX = -\log X$$

Where X is the quantity of interest

Accordingly,

$$pH = -\log[H_3O^+], \qquad pOH = -\log[OH^-]$$

- The relation between these two ion concentration expressed as p-functions is derived from the *K*w expression:

$K_w = [H_3O^+][OH^-]$

$-\log K_w = -\log [H_3O^+][OH^-] = -\log[H_3O^+] + -\log[OH^-]$

$pK_w = pH + pOH$

At 25 °C, $K_w = 1.0 \times 10^{-14}$

14.00 = pH + pOH

For a neutral solution, $[H_3O^+] = [OH^-] = 1.0 \times 10^{-7}$; pH = pOH = 7.0

- Acidic solutions are those H_3O^+ molarities greater than 1.0×10^{-7} *M* and OH^- molarities less than 1.0×10^{-7} *M (*pH < 7.0 and pOH > 7.0*)*
- Basic solutions are those H_3O^+ molarities lesser than 1.0×10^{-7} *M* and OH^- molarities greater than 1.0×10^{-7} *M (*pH > 7.0 and pOH < 7.0*)*
- Air-saturated water has a H_3O^+ concentration of 2.0×10^{-6} *M* caused by the dissolved CO_2. Calculate the pH of the solution at 25 °C.

Solution:

$pH = -\log[H_3O^+] = -\log 2.0 \times 10^{-6} =$ **5.7**

7.6. Relative Strength of Acids and Bases

7.6.1. Relative strength of acids

- We can rank the strengths of acids by the extent to which they ionize in aqueous solution.
- The reaction of an acid with water is given by the general expression:

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

✓ A strong acid yields 100% (or very nearly) of H_3O^+ and A^-

✓ A weak acid gives small amounts of H_3O^+ and A^-

Examples of strong acids and strong bases

6 Strong Acids		6 Strong Bases	
$HClO_4$	perchloric acid	LiOH	lithium hydroxide
HCl	hydrochloric acid	NaOH	sodium hydroxide
HBr	hydrobromic acid	KOH	potassium hydroxide
HI	hydroiodic acid	$Ca(OH)_2$	calcium hydroxide
HNO_3	nitric acid	$Sr(OH)_2$	strontium hydroxide
H_2SO_4	sulfuric acid	$Ba(OH)_2$	barium hydroxide

- The relative strengths of acids may be determined by measuring their equlibrium constants in aqueous solutions.
- In solutions of the same concentration, stronger acids ionize to a greater extent, and therefore yield higher concentrations of hydronium ions than do weaker acids.

The equilibrium constant for an acid is called the **acid-ionization constant, *K*a**.

- For the reaction of an acid HA:

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

Note: We do not include $[H_2O]$ in the equation

✓ The larger the *K*a of an acid, the larger the concentration of H_3O^+ and A^- relative to the concentration of the nonionized acid, HA. Thus,

✓ a stronger acid has a larger ionization constant than does a weaker acid.

Examples:

Acid	Ka(25°C)
HBrO	$2x10^{-9}$
CH3COOH	$1.8x10^{-5}$
HCOOH	$1.8x10^{-4}$
HF	$7.2x10^{-4}$
HCN	$4x10^{-10}$

- Another measure of the strength of an acid is its percent ionization.
- **Percent ionization** of a weak acid: It is the ratio of the concentration of the ionized acid to the initial acid concentration, times 100:

$$\% \text{ ionization} = \frac{[H_3O^+]_{eq}}{[HA]_0} \times 100$$

Note: Percent ionization of a weak acid decreases with increasing acid concentration and increases with increasing ionization constant.

Example:

Calculate the percent ionization of a 0.10 *M* solution of acetic acid with a pH of 2.89.

$$\%ionization = \frac{[H_3O^+]}{[CH_3COOH]} x100$$

pH = -log[H_3O^+] = 2.89

[H_3O^+] = $10^{-2.89}$ = 1.3 x10^{-3}

$$\%ionization = \frac{1.3x10^{-3}}{0.10} x100 = 1.3\%$$

7.6.2. Relative strength of bases

- We can rank the strengths of bases by their tendency to form hydroxide ions in aqueous solution.
- The reaction of a Brønsted-Lowry base with water is given by:

$$B(aq) + H_2O(l) \rightleftharpoons HB^+(aq) + OH^-(aq)$$

➢ Water is the acid that reacts with the base, HB^+ is the conjugate acid of the base B, and OH^- is the conjugate base of water.

- A strong base yields 100% (or very nearly) of OH^- and HB^+ when it reacts with water
- A weak base yields a small proportion of OH^- and HB^+

- We can measure the relative strengths of bases by measuring their **base-ionization constant (*K*b)** in aqueous solutions.
- In solutions of the same concentration, stronger bases ionize to a greater extent, and therefore yield higher hydroxide ion concentrations than do weaker bases.
- A stronger base has a larger ionization constant than does a weaker base.
- For the reaction of a base, B:

$$B(aq) + H_2O(l) \rightleftharpoons HB^+(aq) + OH^-(aq) \quad K_b = \frac{[HB^+][OH^-]}{[B]}$$

Examples:

$CH_3CO_2^-(aq) + H_2O(l) \rightleftharpoons CH_3CO_2H(aq) + OH^-(aq)$ $\quad Kb = 5.6 \times 10^{-10}$

$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$ $\quad Kb = 1.8 \times 10^{-5}$

$(CH_3)_2NH\ (aq) + H_2O(l) \rightleftharpoons (CH_3)_2NH_2^+\ (aq) + OH^-(aq)$ $\quad Kb = 7.4 \times 10^{-4}$

- Percent ionization can be measured for basic solutions

➢ It varies depending on the base ionization constant and the initial concentration of the solution.

$$\%ionization = \frac{[OH^-]_{eq}}{[B]_0} \times 100$$

Relative Strengths of Conjugate Acid-Base Pairs

Consider the ionization reactions for a conjugate acid-base pair, HA – A^-:

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq) \qquad K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

$$A^-(aq) + H_2O(l) \rightleftharpoons OH^-(aq) + HA(aq) \qquad K_b = \frac{[HA][OH]}{[A^-]}$$

Adding these two chemical equations yields the equation for the autoionization for water:

$$2H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

$$K_a \times K_b = \frac{[H_3O^+][A^-]}{[HA]} \times \frac{[HA][OH^-]}{[A^-]} = [H_3O^+][OH^-] = K_w$$

Example:

Ka(CH_3COOH) = 1.8×10^{-5}, Kb of its conjugate base, (CH_3COO^-) = 5.6×10^{-10}.

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq) \qquad K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

$$A^-(aq) + H_2O(l) \rightleftharpoons OH^-(aq) + HA(aq) \qquad K_b = \frac{[HA][OH]}{[A^-]}$$

- Ka x Kb = [H3O+][OH-] = *K*w
- ✓ $K_a \times K_b = (1.8 \times 10^{-5}) \times (5.6 \times 10^{-10}) = 1.0 \times 10^{-14} = K_w$
- The extent to which an acid, HA, donates protons to water molecules depends on the strength of the conjugate base, A^-, of the acid.
- If A^- is a strong base, any protons that are donated to water molecules are recaptured by A^-. Thus
- ✓ there is relatively little A^- and H_3O^+ in solution, and the acid, HA, is weak.
- If A^- is a weak base, water binds the protons more strongly. Thus
- ✓ the solution contains primarily A^- and H_3O^+
- ✓ The acid is strong.

- Strong acids form very weak conjugate bases, and weak acids form stronger conjugate bases
- For strong acid: Ka $\approx \infty$, For the conjugate base Kb = Kw/Ka ≈ 0

Examples:

- $HClO_4$, H_2SO_4, HI, HBr, HCl, HNO_3 undergo complete acid ionization in water

➢ Their conjugate bases (ClO_4^-, HSO_4^- , I^-, Br^-, Cl^-, NO_3^-) do not undergo base ionization in water

✓ The conjugate bases are weaker bases than water

✓ When these acids dissolves in water, their protons are completely transferred to water, the stronger base.

- HS^-, C_2H_5OH, NH_3, CH_4 do not undergo acid ionization in water

➢ They are weaker acids than water

➢ Do not exhibit observable acidic behavior when dissolved in water

➢ Their conjugate bases (S^{2-}, $C_2H_5O^-$, NH_2^-, CH_3^-) are stronger than OH^-

➢ If any conjugate base were formed, it would react with water (accepts a proton) to re-form the acid 100%.

7.6. Relative Strength of Acids and Bases

- Those acids whose strength lie between the hydronium ion and water form conjugate bases that can compete with water for possession of a proton.

➢ Both hydronium ions and nonionized acid molecules are present in equilibrium in a solution

- The extent to which a base forms hydroxide ion in aqueous solution depends on the strength of the base relative to that of the hydroxide ion

➢ A strong base (stronger than OH^-), accepts protons from water to yield 100% of the conjugate acid and hydroxide ion

Examples: S^{2-}, $C_2H_5O^-$, NH_2^-, CH_3^-

➢ Those bases that have strength between water and hydroxide ion accept protons from water, but a mixture of the hydroxide ion and the base results

➢ Bases that are weaker than water show no observable basic behavior in aqueous solution.

Examples: ClO_4^-, HSO_4^-, I^-, Br^-, Cl^-, NO_3^-

Example

Which one is a stronger acid, NH_4^+ or HCN? (Ka (HCN) = 4 x 10^{-10} , Kb (NH_3) = 1.8 x 10^{-5}

Solution:

The conjugate base o f NH_4^+ is NH_3. Therefore,

Ka (NH_4^+) x Kb (NH_3) = Kw =1.0 x 10^{-14} , Ka (NH_4^+) x 1.8 x 10^{-5} = 1.0 x 10^{-14}

Ka (NH_4^+) = 5.6 x10^{-10}

Therefore, NH_4^+ is a stronger acid than HCN.

The Ionization of Weak Acids and Weak Bases

- Consider the ionization of Acetic acid, CH_3CO_2H(weak acid)

$$CH_3CO_2H(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CH_3CO_2^-(aq),$$

At equilibrium:

$$K_a = \frac{[H_3O^+][CH_3CO_2^-]}{[CH_3CO_2H]} = 1.8 \times 10^{-5}$$

Similarly, for the weak base, $(CH_3)_3N$:

$(CH_3)_3N(aq) + H_2O(l) \rightleftharpoons (CH_3)_3\,NH^+(aq) + OH^-(aq)$

At equilibrium:

$$Kb = \frac{[(CH_3)_3\,NH^+][OH^-]}{[(CH_3)_3N]} = 7.4 \times 10^{-5}$$

Example: What is the value of *K*a for acetic acid if at equilibrium the solution contains $[CH_3CO_2H] = 0.0787\ M$ and $[H_3O^+] = [CH_3\,CO_2^-] = 0.00118\ M$?

Solution:

$$CH_3CO_2H\,(aq) + H_2O(l) \rightleftharpoons H_3O^+\,(aq) + CH_3\,CO_2^-\,(aq)$$

$$K_a = \frac{[H_3O^+][CH_3CO_2^-]}{[CH_3CO_2H]} = \frac{(0.00118)(0.00118)}{0.0787} = 1.77 \times 10^{-5}$$

Example:

The pH of a solution of household ammonia, a 0.950 *M* solution of NH_3, is 11.612. What is *K*b for NH_3?

Solution:

$$NH_3\,(aq) + H_2O\,(l) \rightleftharpoons NH_4^+ + OH^-$$

$[NH_4^+] = [OH^-]$; pOH = 14 – pH = 14 – 11.612 = 2.388

$[OH^-] = 10^{-pOH-} = 10^{-2.388} = 4.09 \times 10^{-3}$

$$Kb = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{4.09\ \text{x}\ 10{-}3\ x4.09\ \text{x}\ 10{-}3}{0.950} = \mathbf{1.76 \times 10^{-5}}$$

Example: What is the concentration of hydronium ion, the pH and the % ionization in a 0.534 *M* solution of formic acid?

$HCO_2H(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + HCO_2^-(aq)$ $\quad$ *K*a = 1.8×10^{-4}

Initial concentrations (M): $HCO_2H = 0.534$, $H_3O^+ = HCO_2^- = 0$

Equilibrium concentrations: $HCO_2H = 0.534 - x$, $H_3O^+ = HCO_2^- = x$

$$Ka = \frac{[H_3O^+][HCO_2^-]}{[HCO_2H]} = \frac{x^2}{0.534 - x} = 1.8 \times 10^{-4}$$

*K*a is very small

- The amount reacted (x) is very small when compared with the initial concentration of the acid.
- $x << 0.534$, $0.534 - x \approx 0.534$

$$\frac{x^2}{0.534 - x} = \frac{x^2}{0.534} = 1.8 \times 10^{-4}$$

$$x = 9.8 \times 10^{-3}$$

$[H_3O^+] = x = \mathbf{9.8 \times 10^{-3}\ M}$

$pH = -\log [H_3O^+] = -\log 9.8 \times 10^{-3} = \mathbf{2.01}$

$$\%\ ionization = \frac{[H_3O^+]}{[HCO_2H]} = \frac{9.8 \times 10^{-3}}{0.534} \times 100 = 1.8\%$$

The Relative Strengths of Strong Acids and Bases

- Strong acids, exhibit the same strength in water.
- The water molecule is a strong base compared to the conjugate bases of the strong acids
- Ionization of the strong acids is complete in aqueous solutions.

- In solvents less basic than water, strong acids differ markedly in their tendency to give up a proton to the solvent.

Example:

HCl, HBr, and HI, all exhibit the same strength in water

- In ethanol (a weaker base than water), the acidic strength (the extent of ionization) increases in the order HCl < HBr < HI.
- The inability to discern differences in strength among strong acids dissolved in water is known as the **leveling effect of water**.
- Water also exerts a leveling effect on the strengths of strong bases such as , the oxide ion, O_2^-, and the amide ion, NH_2^-

7.7. Buffer solutions

Buffer Solutions: A solution containing appreciable amounts of a weak conjugate acid-base pair

- Buffer solutions resist a change in pH when small amounts of a strong acid or a strong base is added

A buffer that consists of

a) a weak acid and its salt

Example: acetic acid and sodium acetate ($CH_3COOH + CH_3COONa$)

b) a weak base and its salt

Example: a solution of ammonia and ammonium chloride ($NH_3(aq) + NH_4Cl(aq)$)

- Consider a mixture of roughly equal amounts of acetic acid and sodium acetate.

$$CH_3COOH\ (aq) + H_2O) \rightleftharpoons H_3O^+ + CH_3COO^-$$

- Adding strong base to this solution will neutralize hydronium ion and shift the acetic acid ionization equilibrium to the right, partially restoring the decreased H_3O^+ concentration
- Adding strong acid to this buffer solution will neutralize acetate ion, shifting the above ionization equilibrium right and returning $[H_3O^+]$ to near its original value

- The buffering action of the solution is a result of the added strong acid and base being converted to the weak acid and base that make up the buffer's conjugate pair.
- The weaker acid and base undergo slight ionization, as compared with the complete ionization of the strong acid and base
- The solution pH, therefore, changes much less drastically than it would in an unbuffered solution.

Example: pH changes in buffer and unbuffered solutions

a. Calculate the pH of an acetate buffer that is a mixture with 0.1 M acetic acid and 0.10 M sodium acetate.

b. Calculate the pH after 1.0 mL of 0.10 M NaOH is added to 100 mL of this buffer.

c. For comparison, calculate the pH after 1.0 mL of 0.10 M NaOH is added to 100 mL of a solution of an unbuffered solution with a pH of 4.74.

Solution: $CH_3COOH\ (aq) + H_2O) \rightleftharpoons H_3O^+ + CH_3COO^-$ $Ka = 1.77 \times 10^{-5}$

a) Initial concentrations (M): $CH_3COOH = 0.1$, $H_3O^+ = 0$ $CH_3COO^- = 0.10$

Change (M): $CH_3COOH = -x$, $H_3O^+ = +x$ $CH_3COO^- = +x$

Equilibrium concentration(M): $CH_3COOH = 0.1 - x$, $H_3O^+ = x$ $CH_3COO^- = 0.10 + x$

$$1.77 \times 10^{-5} = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]} = \frac{x \times (0.1 + x)}{0.1 - x}$$

Assuming $X << 0.1$, $x = 1.77 \times 10^{-5}$ M

$[H_3O^+] = 1.77 \times 10^{-5}$ M, pH = -log 1.77×10^{-5} = **4.75**

b) In 1.0 mL of 0.10 M NaOH, there are 1.0×10^{-4} mol NaOH

In 100 mL of 0.1 M CH_3COOH, there are 0.01 mol of CH_3COOH

1.0×10^{-4} mol of CH_3COOH reacts with 1.0×10^{-4} mol NaOH to produce 1.0×10^{-4} mol of $NaCH_3COO$

Amount of CH_3COOH remaining = 9.9×10^{-3} mol

Final number of moles of acetate ion becomes $1.0 \times 10^{-4} + 0.01 = 1.01 \times 10^{-2}$

Total volume = 100 mL + 1 mL = 101 mL = 0.101 L

$[CH_3COOH] = 9.9 \times 10^{-3}$ mol/ 0.101 L = 0.098 M

$[NaCH_3COO] = 1.01 \times 10^{-2}$/ 0.101 L = 0.1M

$$1.77 \times 10^{-5} = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]} = \frac{[H_3O^+] \times 0.1}{0.098}$$

$[H_3O^+]$ = 1.73 x 10^{-5} M, **pH = 4.76**

- The change in pH before and after the addition of 0.1 M NaOH is very small (0.01)

c) pH = 4.74, $[H_3O^+]$ = 1.82 x 10^{-5} M

1.0 mL of 0.10 M NaOH = 1.0 x 10^{-4} mol NaOH = 1.0 x 10^{-4} mol OH^-

100 mL of 1.73 x 10^{-5} M H_3O^+ = 1.82 x 10^{-6} mol

The stoichiometric reaction ratio between OH^- and H_3O^+ is 1: 1

Therefore amount of OH^- remaining after the reaction is:

= 1.0 x 10^{-4} mol - 1.82 x 10^{-6} mol = 9.82 x 10^{-5} mol

Total volume = 0.101 L

$[OH^-]$ = 9.82 x 10^{-5} mol/ 0.101 L = 9.7 x 10^{-4} M

pOH = 3.01, **pH = 10.99**

- In this unbuffered solution, addition of the base results in a significant rise in pH (from 4.74 to 10.99)